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COLLEGE OF ENGINEERING
NO - 1, Kennedy Road, Near R.T.O Pune - 411001

Experiment -01.

- AIM: Linear regression by using Deep Neural Network - Implement boston housing price prediction problem by linear regression using deep Neural network. Use boston House price prediction dataset.

- THEORY:

(1) Introduction.

- House price prediction is a classic regression problem in machine Learning where the objective is to estimate the monetary value of a house based on various socio-economic and geographical factors. Traditionally, this problem has been solved using linear regression, where a single linear equation models the relationship between input features and output price.

- With the advancement of deep learning, Deep Neural Networks (DNNs) have emerged as a powerful alternative for regression tasks.
- Even when the underlying relationship is mostly linear, neural networks provide flexibility, robustness to noise and better scalability when the data size or feature Complexity increases.

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- In this experiment, the Boston Housing Price dataset is used to implement linear regression using a deep neural network, demonstrating how a neural network can be configured to behave as a linear regression model and extended to a deeper architecture for improved learning capability.

(2) Overview of Boston Housing Dataset -

- The boston housing price dataset is a well known benchmark dataset originally collected by the U.S. Census service. It contains information about housing values in the suburbs of boston.

- Dataset characteristics -
  - Total instances : 506.
  - Number of features : 13.
  - Target variable: Median value of owner occupied homes (MEDV)

- Input features description -

1. CRIM - Per capita crime rate.
2. ZN - Proportion of residential land zoned for large clots.
3. INDUS - Proportion of non retail business acres.
4. CHAS - Charles river dummy variable.
5. NOX - Nitric oxide concentration.
6. RM - Average number of rooms per dwelling.

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(3) Target variable - MEDV : Median house price (in $1000s).

(4) Linear regression concept :-

It attempts to model the relationship between independent variables X and dependent variable y using a linear equation.

$$y = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b.$$

Where,

- w_i are the weights.
- b is the bias.
- x_i are the input features.

(5) Why use a Deep Neural Network for Linear regression?

→ Although linear regression can be solved analytically, implementing it using a deep neural network offers several advantages:

- Unified framework for both regression and classification.
- Easy extension to non linear regression.
- Automatic gradient composition using back propagation
- Scalability to large datasets.
- Better generalization when regulalization techniques are applied.

When a neural network uses no activation function (or linear activation) in the output layer, it effectively performs linear regression.

(6) Deep Neural Network architecture for regression -

Model structure

A typical DNN used for linear regression consists of -

1. Input layer - Accepts 13 numerical features from Dataset.

2. Hidden Layers -
   - One or more fully connected (Dense) layers.
   - Use activation functions such as ReLU.
   - Learn higher level feature interactions.

3. Output layer -
   - Single Neuron.
   - Linear activation.
   - Outputs predicted house price.

Example architecture -

- Input layer : 13 neurons.
- Hidden layer 1 : 64 neurons (ReLU).
- Hidden layer 2 : 32 neurons (ReLU).
- Output layer : 1 neurons (Linear)

(7). Model Training

- Loss Function - For regression problems, Mean squared error (MSE) is commonly used :

$$MSE = \frac{1}{n} \sum (y_{true} - y_{pred})^2$$

Optimizer:
- Gradient descent / Adam optimizer.
- Updates weights using backpropagation.

- Training process -
  1. Forward pass to compute predictions.
  2. Compute loss.
  3. Backpropagate error.
  4. Update weights.
  5. Repeat for multiple epochs.

(8). Results and Observations -

- The deep neural network successfully learns the mapping between input features and house prices.
- Feature scaling significantly improves convergence speed.
- Increasing depth allows the model to capture complex interactions.
- Linear activation in the output layer ensures continuous valued predictions.
- Overfitting can occur if the network is too deep, requiring regularization.

(9) Advantages of using DNN for Linear regression -

- Handles multivariate data efficiently.
- Easily extendable to non linear problems.
- Supports Large scale datasets.
- Integrates well with modern ML pipelines.

(10) Limitations -

- Requires more computation than traditional linear regression.
- Needs careful training of hyperparameters.
- Boston dataset is relatively small, limiting deep architecture.

(11). Applications -

- Real estate price prediction.
- Financial forecasting.
- Demand estimation.
- Risk analysis.
- Economic modeling.

- CONCLUSION : Thus we implemented Deep Neural network on Boston dataset.

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